oyee-attrition-case-study-mainfile

April 18, 2024

1 Employee Attrition Case Study

1.1 Section 01

1.1.1 Overall Business Objective:

The objective is to investigate and understand the factors that lead to employee attrition in the company. By analyzing the data, the goal is to identify patterns, trends, and insights that can help in reducing attrition and improving employee retention. ### Understanding of the Problem: The business analyst will align their understanding of the client's requirements with the problem of employee attrition. This ensures that the analysis is focused on addressing the specific needs of the client. ### Approach: The business analyst will follow a well-defined plan to address the problem. The high-level approach may include data health review, exploratory data analysis (EDA), and KPI/metric-based questions to generate insights and provide actionable recommendations.

[1]:
```
# importing all important package....
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
import warnings
warnings.filterwarnings('ignore')
from scipy.stats import norm, skew
from scipy import stats
import statsmodels.api as sm
import plotly.express as px
```

[2]:
```
#load data into pandas dataframe..
df = pd.read_excel('Employee Attrition Data 0227.xlsx','Data')
df.head(5)
```

[2]:
```
   Age Attrition     BusinessTravel              Department  DistanceFromHome  \
0   51        No      Travel_Rarely                   Sales                 6
1   31       Yes  Travel_Frequently  Research & Development                10
2   32        No  Travel_Frequently  Research & Development                17
3   38        No         Non-Travel  Research & Development                 2
4   32        No      Travel_Rarely  Research & Development                10

   Education EducationField  EmployeeID  Gender  JobLevel  … MaritalStatus  \
```

```
0          2  Life Sciences           1  Female         1  …       Married
1          1  Life Sciences           2  Female         1  …        Single
2          4          Other           3    Male         4  …       Married
3          5  Life Sciences           4    Male         3  …       Married
4          1        Medical           5    Male         1  …        Single

  MonthlyIncome  NumCompaniesWorked  PercentSalaryHike  StockOptionLevel  \
0        131160                 1.0                 11                 0
1         41890                 0.0                 23                 1
2        193280                 1.0                 15                 3
3         83210                 3.0                 11                 3
4         23420                 4.0                 12                 2

   TotalWorkingYears  TrainingTimesLastYear  YearsAtCompany  \
0               1.0                      6               1
1               6.0                      3               5
2               5.0                      2               5
3              13.0                      5               8
4               9.0                      2               6

   YearsSinceLastPromotion  YearsWithCurrManager
0                        0                     0
1                        1                     4
2                        0                     3
3                        7                     5
4                        0                     4

[5 rows x 21 columns]
```

1.2 Section 02: Data Health Review

[3]:
```
# information of the dataset for checking whether the columns are in the right␣
↪format is their any null values present in columns...
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 4410 entries, 0 to 4409
Data columns (total 21 columns):
 #   Column                   Non-Null Count  Dtype
---  ------                   --------------  -----
 0   Age                      4410 non-null   int64
 1   Attrition                4410 non-null   object
 2   BusinessTravel           4410 non-null   object
 3   Department               4410 non-null   object
 4   DistanceFromHome         4410 non-null   int64
 5   Education                4410 non-null   int64
 6   EducationField           4410 non-null   object
```

```
 7   EmployeeID                4410 non-null   int64
 8   Gender                    4410 non-null   object
 9   JobLevel                  4410 non-null   int64
 10  JobRole                   4410 non-null   object
 11  MaritalStatus             4410 non-null   object
 12  MonthlyIncome             4410 non-null   int64
 13  NumCompaniesWorked        4391 non-null   float64
 14  PercentSalaryHike         4410 non-null   int64
 15  StockOptionLevel          4410 non-null   int64
 16  TotalWorkingYears         4401 non-null   float64
 17  TrainingTimesLastYear     4410 non-null   int64
 18  YearsAtCompany            4410 non-null   int64
 19  YearsSinceLastPromotion   4410 non-null   int64
 20  YearsWithCurrManager      4410 non-null   int64
dtypes: float64(2), int64(12), object(7)
memory usage: 723.6+ KB
```

1.3 Dealing with null values

[4]:
```python
#  checking the sum of null values in every column...
totalnull_val = df.isnull().sum()
totalnull_val
```

[4]:
```
Age                         0
Attrition                   0
BusinessTravel              0
Department                  0
DistanceFromHome            0
Education                   0
EducationField              0
EmployeeID                  0
Gender                      0
JobLevel                    0
JobRole                     0
MaritalStatus               0
MonthlyIncome               0
NumCompaniesWorked         19
PercentSalaryHike           0
StockOptionLevel            0
TotalWorkingYears           9
TrainingTimesLastYear       0
YearsAtCompany              0
YearsSinceLastPromotion     0
YearsWithCurrManager        0
dtype: int64
```

[5]:
```
# calculating the percentage of null values for every individual column….
percentnull_val = (totalnull_val/df.shape[0])*100
percentnull_val
```

[5]:
```
Age                        0.000000
Attrition                  0.000000
BusinessTravel             0.000000
Department                 0.000000
DistanceFromHome           0.000000
Education                  0.000000
EducationField             0.000000
EmployeeID                 0.000000
Gender                     0.000000
JobLevel                   0.000000
JobRole                    0.000000
MaritalStatus              0.000000
MonthlyIncome              0.000000
NumCompaniesWorked         0.430839
PercentSalaryHike          0.000000
StockOptionLevel           0.000000
TotalWorkingYears          0.204082
TrainingTimesLastYear      0.000000
YearsAtCompany             0.000000
YearsSinceLastPromotion    0.000000
YearsWithCurrManager       0.000000
dtype: float64
```

1.3.1 We can observe that NumCompaniesWorked and TotalWorkingYears have 0.43% & 0.20% percent null values respectively Hence our decision of either drop the Null values or imputing them. As the percent null values were very low So we can go ahead and drop null values.

[6]:
```
# calculating the total no of null values for complete table ….
totalnull_val = df.isnull().sum().sum()
totalnull_val
```

[6]: 28

[7]:
```
# calculating the percent of total no of null values for complete table ….
percentnull_val = (totalnull_val/df.shape[0])*100
percentnull_val
```

[7]: 0.6349206349206349

1.3.2 Hence 0.63% of values were only affected. So, we can go ahead and drop null values.

[8]:
```
df.shape # With null values
```

[8]:
```
(4410, 21)
```

[9]:
```
# droping null values
df = df.dropna()
```

[10]:
```
df.shape #After removing null values
```

[10]:
```
(4382, 21)
```

1.4 Dealing with datatypes....

[11]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Int64Index: 4382 entries, 0 to 4408
Data columns (total 21 columns):
 #   Column                   Non-Null Count  Dtype
---  ------                   --------------  -----
 0   Age                      4382 non-null   int64
 1   Attrition                4382 non-null   object
 2   BusinessTravel           4382 non-null   object
 3   Department               4382 non-null   object
 4   DistanceFromHome         4382 non-null   int64
 5   Education                4382 non-null   int64
 6   EducationField           4382 non-null   object
 7   EmployeeID               4382 non-null   int64
 8   Gender                   4382 non-null   object
 9   JobLevel                 4382 non-null   int64
 10  JobRole                  4382 non-null   object
 11  MaritalStatus            4382 non-null   object
 12  MonthlyIncome            4382 non-null   int64
 13  NumCompaniesWorked       4382 non-null   float64
 14  PercentSalaryHike        4382 non-null   int64
 15  StockOptionLevel         4382 non-null   int64
 16  TotalWorkingYears        4382 non-null   float64
 17  TrainingTimesLastYear    4382 non-null   int64
 18  YearsAtCompany           4382 non-null   int64
 19  YearsSinceLastPromotion  4382 non-null   int64
 20  YearsWithCurrManager     4382 non-null   int64
dtypes: float64(2), int64(12), object(7)
memory usage: 753.2+ KB
```

1.4.1 As we can see the EmployeeID, Education, JobLevel, NumCompaniesWorked, TotalWorkingYears columns are not in the correct format changing them in to required datatype....

[12]:
```
# convert datatypes of specific columns
df['EmployeeID'] = df['EmployeeID'].astype(object)
df['Education'] = df['Education'].astype(object)
df['JobLevel'] = df['JobLevel'].astype(object)
df['NumCompaniesWorked'] = df['NumCompaniesWorked'].astype(int)
df['TotalWorkingYears'] = df['TotalWorkingYears'].astype(int)
```

[13]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Int64Index: 4382 entries, 0 to 4408
Data columns (total 21 columns):
 #   Column                    Non-Null Count  Dtype
---  ------                    --------------  -----
 0   Age                       4382 non-null   int64
 1   Attrition                 4382 non-null   object
 2   BusinessTravel            4382 non-null   object
 3   Department                4382 non-null   object
 4   DistanceFromHome          4382 non-null   int64
 5   Education                 4382 non-null   object
 6   EducationField            4382 non-null   object
 7   EmployeeID                4382 non-null   object
 8   Gender                    4382 non-null   object
 9   JobLevel                  4382 non-null   object
 10  JobRole                   4382 non-null   object
 11  MaritalStatus             4382 non-null   object
 12  MonthlyIncome             4382 non-null   int64
 13  NumCompaniesWorked        4382 non-null   int32
 14  PercentSalaryHike         4382 non-null   int64
 15  StockOptionLevel          4382 non-null   int64
 16  TotalWorkingYears         4382 non-null   int32
 17  TrainingTimesLastYear     4382 non-null   int64
 18  YearsAtCompany            4382 non-null   int64
 19  YearsSinceLastPromotion   4382 non-null   int64
 20  YearsWithCurrManager      4382 non-null   int64
dtypes: int32(2), int64(9), object(10)
memory usage: 718.9+ KB
```

[14]:
```
# check the no of unique values in every column for better data understanding
df.nunique()
```

[14]:
```
Age                          43
Attrition                     2
BusinessTravel                3
```

```
Department                    3
DistanceFromHome             29
Education                     5
EducationField                6
EmployeeID                 4382
Gender                        2
JobLevel                      5
JobRole                       9
MaritalStatus                 3
MonthlyIncome              1349
NumCompaniesWorked           10
PercentSalaryHike            15
StockOptionLevel              4
TotalWorkingYears            40
TrainingTimesLastYear         7
YearsAtCompany               37
YearsSinceLastPromotion      16
YearsWithCurrManager         18
dtype: int64
```

[15]:
```python
# Looking at all the unique values in features

for i in df.columns:
    print("Unique values in",'',i)
    unique_vals = df[i].unique()
    print(unique_vals)
    print(50*'*')
```

```
Unique values in  Age
[51 31 32 38 46 28 29 25 45 36 55 47 37 21 35 26 50 53 44 49 42 18 41 39
 58 33 43 52 27 30 54 40 23 48 57 34 24 22 56 60 19 20 59]
**************************************************
Unique values in  Attrition
['No' 'Yes']
**************************************************
Unique values in  BusinessTravel
['Travel_Rarely' 'Travel_Frequently' 'Non-Travel']
**************************************************
Unique values in  Department
['Sales' 'Research & Development' 'Human Resources']
**************************************************
Unique values in  DistanceFromHome
[ 6 10 17  2  8 11 18  1  7 28 14  3 16  9  5  4 20 29 15 13 24 19 22 25
 21 26 27 12 23]
**************************************************
Unique values in  Education
[2 1 4 5 3]
```

```
**************************************************
Unique values in  EducationField
['Life Sciences' 'Other' 'Medical' 'Marketing' 'Technical Degree'
 'Human Resources']
**************************************************
Unique values in  EmployeeID
[1 2 3 … 4407 4408 4409]
**************************************************
Unique values in  Gender
['Female' 'Male']
**************************************************
Unique values in  JobLevel
[1 4 3 2 5]
**************************************************
Unique values in  JobRole
['Healthcare Representative' 'Research Scientist' 'Sales Executive'
 'Human Resources' 'Research Director' 'Laboratory Technician'
 'Manufacturing Director' 'Sales Representative' 'Manager']
**************************************************
Unique values in  MaritalStatus
['Married' 'Single' 'Divorced']
**************************************************
Unique values in  MonthlyIncome
[131160  41890 193280 …  48050  44340  83800]
**************************************************
Unique values in  NumCompaniesWorked
[1 0 3 4 2 7 9 5 6 8]
**************************************************
Unique values in  PercentSalaryHike
[11 23 15 12 13 20 22 21 17 14 16 18 19 24 25]
**************************************************
Unique values in  StockOptionLevel
[0 1 3 2]
**************************************************
Unique values in  TotalWorkingYears
[ 1  6  5 13  9 28 10 21 16 37  7  3 15  8 12 17 19 22  2  4 23  0 11 24
 25 20 14 26 18 30 36 31 33 32 34 40 29 35 27 38]
**************************************************
Unique values in  TrainingTimesLastYear
[6 3 2 5 4 0 1]
**************************************************
Unique values in  YearsAtCompany
[ 1  5  8  6  7  0  9 20 15 36 10  3 17  2  4 11 22 18 13 24 21 16 25 29
 27 14 31 32 34 26 12 19 33 30 23 37 40]
**************************************************
Unique values in  YearsSinceLastPromotion
[ 0  1  7  4 10  9  6  3  5  2  8 11 13 12 15 14]
**************************************************
```

```
Unique values in  YearsWithCurrManager
[ 0  4  3  5  7  8 10 11 13  9  1  2  6 12 17 16 15 14]
**************************************************
```

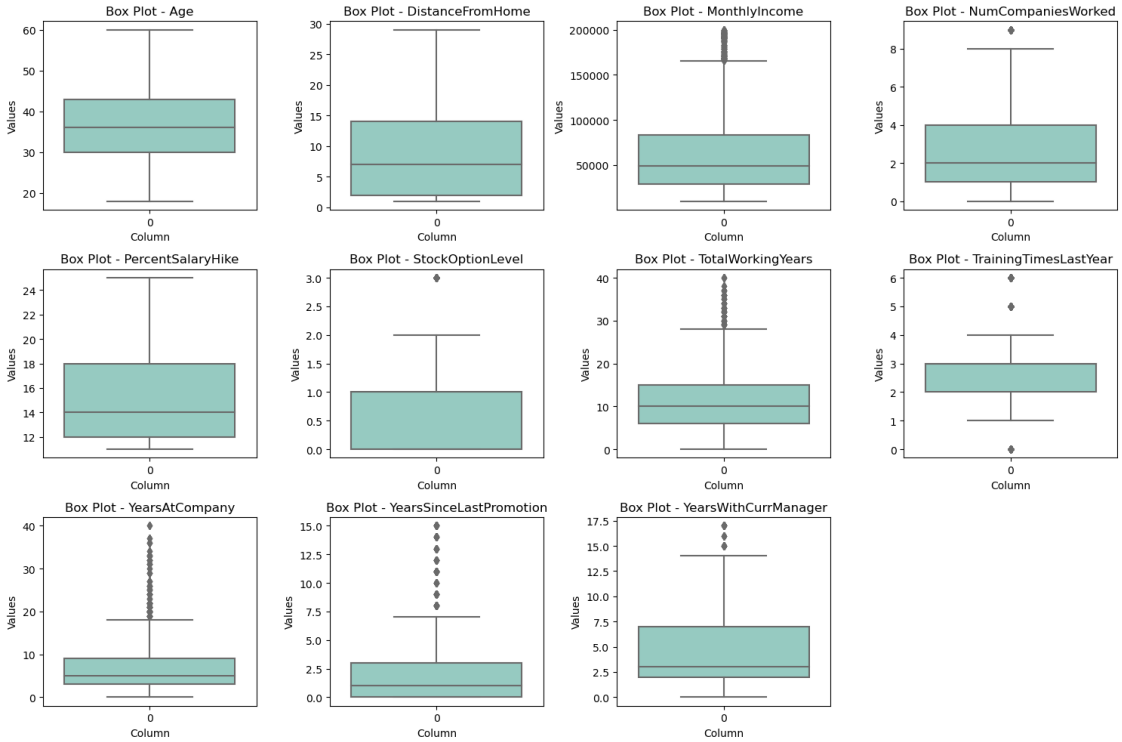
1.5 Dealing with Out layers....

[16]:
```python
numerical_columns = df.select_dtypes(include='number')
numerical_columns.head()
```

[16]:
```
   Age  DistanceFromHome  MonthlyIncome  NumCompaniesWorked  \
0   51                 6         131160                   1
1   31                10          41890                   0
2   32                17         193280                   1
3   38                 2          83210                   3
4   32                10          23420                   4

   PercentSalaryHike  StockOptionLevel  TotalWorkingYears  \
0                 11                 0                  1
1                 23                 1                  6
2                 15                 3                  5
3                 11                 3                 13
4                 12                 2                  9

   TrainingTimesLastYear  YearsAtCompany  YearsSinceLastPromotion  \
0                      6               1                        0
1                      3               5                        1
2                      2               5                        0
3                      5               8                        7
4                      2               6                        0

   YearsWithCurrManager
0                     0
1                     4
2                     3
3                     5
4                     4
```

[17]:
```python
# Create box plots for all columns one after the other
plt.figure(figsize=(15, 10))
for i, col in enumerate(numerical_columns.columns, 1):
    plt.subplot(3, 4, i)
    sns.boxplot(data=numerical_columns[col], palette='Set3')
    plt.title(f"Box Plot - {col}")
    plt.xlabel("Column")
    plt.ylabel("Values")
    plt.tight_layout()
```

```
plt.show()
```



In this code, we use sns.stripplot to create interactive strip plots for each column. The data=df parameter specifies the DataFrame, y=col sets the column on the y-axis, and color='blue' defines the color of the data points in the plot.

This is because swarmplots are slower to render on Jupyter notebooks.

1.6 Dealing with duplicate values ....

[18]:
```
df.duplicated().sum()
```

[18]:
```
0
```

there are no duplicate values in the column

1.7 Dealing with data cleaning ....

[19]:
```
df.columns
```

[19]:
```
Index(['Age', 'Attrition', 'BusinessTravel', 'Department', 'DistanceFromHome',
       'Education', 'EducationField', 'EmployeeID', 'Gender', 'JobLevel',
       'JobRole', 'MaritalStatus', 'MonthlyIncome', 'NumCompaniesWorked',
       'PercentSalaryHike', 'StockOptionLevel', 'TotalWorkingYears',
```

```
       'TrainingTimesLastYear', 'YearsAtCompany', 'YearsSinceLastPromotion',
       'YearsWithCurrManager'],
      dtype='object')
```

the column names should be changed unable to understand

[20]:
```python
#columns names should be change
new_column_names = {
    'BusinessTravel': 'Business_Travel',
    'DistanceFromHome': 'Distance_From_Home',
    'EducationField': 'Education_Field',
    'EmployeeID': 'Employee_ID',
    'JobLevel': 'Job_Level',
    'JobRole': 'Job_Role',
    'MaritalStatus': 'Marital_Status',
    'MonthlyIncome': 'Monthly_Income',
    'NumCompaniesWorked': 'Num_Companies_Worked',
    'PercentSalaryHike': 'Percent_Salary_Hike',
    'StockOptionLevel': 'Stock_Option_Level',
    'TotalWorkingYears': 'Total_Working_Years',
    'TrainingTimesLastYear': 'Training_Times_Last_Year',
    'YearsAtCompany': 'Years_At_Company',
    'YearsSinceLastPromotion': 'Years_Since_Last_Promotion',
    'YearsWithCurrManager': 'Years_With_Curr_Manager'
}
df = df.rename(columns=new_column_names)
df.head(10)
```

[20]:
```
   Age Attrition    Business_Travel              Department  \
0   51        No      Travel_Rarely                   Sales
1   31       Yes  Travel_Frequently  Research & Development
2   32        No  Travel_Frequently  Research & Development
3   38        No         Non-Travel  Research & Development
4   32        No      Travel_Rarely  Research & Development
5   46        No      Travel_Rarely  Research & Development
6   28       Yes      Travel_Rarely  Research & Development
7   29        No      Travel_Rarely  Research & Development
8   31        No      Travel_Rarely  Research & Development
9   25        No         Non-Travel  Research & Development

   Distance_From_Home Education Education_Field Employee_ID  Gender Job_Level  \
0                   6         2   Life Sciences           1  Female         1
1                  10         1   Life Sciences           2  Female         1
2                  17         4           Other           3    Male         4
3                   2         5   Life Sciences           4    Male         3
4                  10         1         Medical           5    Male         1
5                   8         3   Life Sciences           6  Female         4
```

6	11	2	Medical	7	Male	2
7	18	3	Life Sciences	8	Male	2
8	1	3	Life Sciences	9	Male	3
9	7	4	Medical	10	Female	4

	…	Marital_Status	Monthly_Income	Num_Companies_Worked	\
0	…	Married	131160	1	
1	…	Single	41890	0	
2	…	Married	193280	1	
3	…	Married	83210	3	
4	…	Single	23420	4	
5	…	Married	40710	3	
6	…	Single	58130	2	
7	…	Married	31430	2	
8	…	Married	20440	0	
9	…	Divorced	134640	1	

	Percent_Salary_Hike	Stock_Option_Level	Total_Working_Years	\
0	11	0	1	
1	23	1	6	
2	15	3	5	
3	11	3	13	
4	12	2	9	
5	13	0	28	
6	20	1	5	
7	22	3	10	
8	21	0	10	
9	13	1	6	

	Training_Times_Last_Year	Years_At_Company	Years_Since_Last_Promotion	\
0	6	1	0	
1	3	5	1	
2	2	5	0	
3	5	8	7	
4	2	6	0	
5	5	7	7	
6	2	0	0	
7	2	0	0	
8	2	9	7	
9	2	6	1	

	Years_With_Curr_Manager
0	0
1	4
2	3
3	5
4	4

```
5                        7
6                        0
7                        0
8                        8
9                        5

[10 rows x 21 columns]
```

2 Generate Extended Data Dictionary (EDD) of the provided dataset to help comment on data quality

[21]:
```
summary_df = pd.DataFrame({
    'min': df.min(),
    'max': df.max(),
    'median': df.median(),
    'mean': df.mean(),
    'std': df.std(),
    '5%ile': df.quantile(0.05),
    '10%ile': df.quantile(0.10),
    '25%ile': df.quantile(0.25),
    '50%ile': df.quantile(0.50),
    '75%ile': df.quantile(0.75),
    '90%ile': df.quantile(0.90),
    '95%ile': df.quantile(0.95),
    'count': df.count(),
    'na': df.isnull().sum()})

summary_df.head(21)
```

[21]:
```
                                                 min                   max  \
Age                                               18                    60
Attrition                                         No                   Yes
Business_Travel                           Non-Travel         Travel_Rarely
Department                            Human Resources                 Sales
Distance_From_Home                                 1                    29
Education                                          1                     5
Education_Field                       Human Resources      Technical Degree
Employee_ID                                        1                  4409
Gender                                        Female                  Male
Job_Level                                          1                     5
Job_Role                   Healthcare Representative  Sales Representative
Marital_Status                               Divorced                Single
Monthly_Income                                  10090                199990
Num_Companies_Worked                                0                     9
Percent_Salary_Hike                                11                    25
Stock_Option_Level                                  0                     3
```

Total_Working_Years	0	40
Training_Times_Last_Year	0	6
Years_At_Company	0	40
Years_Since_Last_Promotion	0	15
Years_With_Curr_Manager	0	17

	median	mean	std	5%ile	\
Age	36.0	36.933364	9.137272	24.0	
Attrition	NaN	NaN	NaN	NaN	
Business_Travel	NaN	NaN	NaN	NaN	
Department	NaN	NaN	NaN	NaN	
Distance_From_Home	7.0	9.198996	8.105396	1.0	
Education	3.0	2.912369	1.024728	NaN	
Education_Field	NaN	NaN	NaN	NaN	
Employee_ID	2208.5	2207.804884	1271.688783	NaN	
Gender	NaN	NaN	NaN	NaN	
Job_Level	2.0	2.063898	1.106115	NaN	
Job_Role	NaN	NaN	NaN	NaN	
Marital_Status	NaN	NaN	NaN	NaN	
Monthly_Income	49190.0	65061.702419	47142.310175	20970.0	
Num_Companies_Worked	2.0	2.693291	2.497832	0.0	
Percent_Salary_Hike	14.0	15.210634	3.663007	11.0	
Stock_Option_Level	1.0	0.794614	0.852397	0.0	
Total_Working_Years	10.0	11.290278	7.785717	1.0	
Training_Times_Last_Year	3.0	2.798266	1.289402	1.0	
Years_At_Company	5.0	7.010497	6.129351	1.0	
Years_Since_Last_Promotion	1.0	2.191693	3.224994	0.0	
Years_With_Curr_Manager	3.0	4.126198	3.569674	0.0	

	10%ile	25%ile	50%ile	75%ile	90%ile	\
Age	26.0	30.0	36.0	43.0	50.0	
Attrition	NaN	NaN	NaN	NaN	NaN	
Business_Travel	NaN	NaN	NaN	NaN	NaN	
Department	NaN	NaN	NaN	NaN	NaN	
Distance_From_Home	1.0	2.0	7.0	14.0	23.0	
Education	NaN	NaN	NaN	NaN	NaN	
Education_Field	NaN	NaN	NaN	NaN	NaN	
Employee_ID	NaN	NaN	NaN	NaN	NaN	
Gender	NaN	NaN	NaN	NaN	NaN	
Job_Level	NaN	NaN	NaN	NaN	NaN	
Job_Role	NaN	NaN	NaN	NaN	NaN	
Marital_Status	NaN	NaN	NaN	NaN	NaN	
Monthly_Income	23140.0	29110.0	49190.0	83790.0	138260.0	
Num_Companies_Worked	0.0	1.0	2.0	4.0	7.0	
Percent_Salary_Hike	11.0	12.0	14.0	18.0	21.0	
Stock_Option_Level	0.0	0.0	1.0	1.0	2.0	
Total_Working_Years	3.0	6.0	10.0	15.0	23.0	

```
Training_Times_Last_Year        2.0      2.0      3.0      3.0       5.0
Years_At_Company                1.0      3.0      5.0      9.0      15.0
Years_Since_Last_Promotion      0.0      0.0      1.0      3.0       7.0
Years_With_Curr_Manager         0.0      2.0      3.0      7.0       9.0

                              95%ile  count  na
Age                             54.0   4382   0
Attrition                        NaN   4382   0
Business_Travel                  NaN   4382   0
Department                       NaN   4382   0
Distance_From_Home              26.0   4382   0
Education                        NaN   4382   0
Education_Field                  NaN   4382   0
Employee_ID                      NaN   4382   0
Gender                           NaN   4382   0
Job_Level                        NaN   4382   0
Job_Role                         NaN   4382   0
Marital_Status                   NaN   4382   0
Monthly_Income              178560.0   4382   0
Num_Companies_Worked             8.0   4382   0
Percent_Salary_Hike             22.0   4382   0
Stock_Option_Level               3.0   4382   0
Total_Working_Years             28.0   4382   0
Training_Times_Last_Year         5.0   4382   0
Years_At_Company                20.0   4382   0
Years_Since_Last_Promotion       9.0   4382   0
Years_With_Curr_Manager         10.0   4382   0
```

From the above table, we can observe the Extended Data Dictionary of every column and get insights for every column…..

2.1 Section 03: Exploratory Data Analysis

2.2 Attrition Happened

[22]:
```
# Visualizing target variable classes and its distribution among the dataset
fig = px.pie(df, names = 'Attrition', title = 'Target Variable: Attrition',␣
 ↪template = 'plotly_dark')
fig.update_traces(rotation=90, pull = [0.1], textinfo = "percent+label")
fig.show()
```

2.3 Univariate Analysis

2.3.1 Numerical Features

[23]:
```
Attrition = df.query("Attrition == 'Yes' ")
Attrition
```

[23]:
```
      Age Attrition    Business_Travel              Department  \
1      31       Yes  Travel_Frequently  Research & Development
6      28       Yes      Travel_Rarely  Research & Development
13     47       Yes         Non-Travel  Research & Development
28     44       Yes  Travel_Frequently  Research & Development
30     26       Yes      Travel_Rarely  Research & Development
…    …        …                 …                       …
4381   29       Yes      Travel_Rarely  Research & Development
4386   33       Yes      Travel_Rarely                   Sales
4388   33       Yes      Travel_Rarely                   Sales
4391   32       Yes      Travel_Rarely                   Sales
4402   37       Yes  Travel_Frequently                   Sales

      Distance_From_Home Education Education_Field Employee_ID  Gender  \
1                     10         1   Life Sciences           2  Female
6                     11         2         Medical           7    Male
13                     1         1         Medical          14    Male
28                     1         2         Medical          29    Male
30                     4         3         Medical          31    Male
…                    …        …               …         …      …
4381                   7         1   Life Sciences        4382  Female
4386                  11         4       Marketing        4387    Male
4388                   1         3   Life Sciences        4389    Male
4391                  23         1   Life Sciences        4392    Male
4402                   2         3       Marketing        4403    Male

     Job_Level  … Marital_Status Monthly_Income  Num_Companies_Worked  \
1            1  …         Single          41890                     0
6            2  …         Single          58130                     2
13           1  …        Married          57620                     1
28           2  …       Divorced         103330                     3
30           3  …       Divorced          68540                     2
…          …  …              …              …                     …
4381         1  …         Single          21800                     1
4386         1  …        Married          71400                     5
4388         2  …        Married          51470                     7
4391         3  …         Single          24680                     0
4402         1  …       Divorced          40010                     6

      Percent_Salary_Hike  Stock_Option_Level  Total_Working_Years  \
1                      23                   1                    6
```

```
6                       20                    1                     5
13                      11                    2                    10
28                      14                    1                    19
30                      11                    0                     5
…                    …                  …                    …
4381                    21                    0                     4
4386                    21                    0                     8
4388                    11                    0                    13
4391                    11                    0                     4
4402                    11                    1                    17

      Training_Times_Last_Year  Years_At_Company  Years_Since_Last_Promotion  \
1                            3                 5                           1
6                            2                 0                           0
13                           4                10                           9
28                           2                 1                           0
30                           5                 3                           0
…                          …                 …                           …
4381                         2                 4                           0
4386                         2                 5                           0
4388                         2                 9                           1
4391                         2                 3                           1
4402                         2                 1                           0

      Years_With_Curr_Manager
1                           4
6                           0
13                          9
28                          0
30                          2
…                         …
4381                        1
4386                        4
4388                        7
4391                        2
4402                        0

[705 rows x 21 columns]
```

[24]:
```python
# Creating a function to plot histograms on numeric features
def numeric_plot(i):
    fig = px.histogram(Attrition, x = Attrition[i], nbins  = 20, template =␣
 ↪'plotly_dark',
                       title = f'Attrition in {i}')
    fig.show()
```

[25]:
```
numeric_plot('Age')
numeric_plot('Distance_From_Home')
numeric_plot('Education')
numeric_plot('Job_Level')
numeric_plot('Monthly_Income')
numeric_plot('Num_Companies_Worked')
numeric_plot('Percent_Salary_Hike')
numeric_plot('Stock_Option_Level')
numeric_plot('Total_Working_Years')
numeric_plot('Training_Times_Last_Year')
numeric_plot('Years_At_Company')
numeric_plot('Years_Since_Last_Promotion')
numeric_plot('Years_With_Curr_Manager')
```

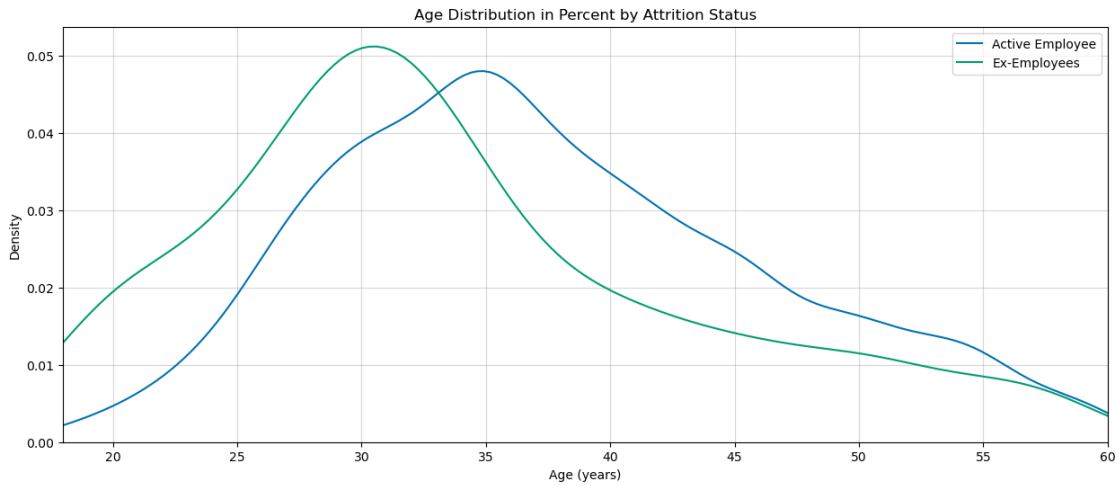
A few observations can be made based on the information and histograms for numerical features:

1. Many histograms are tail-heavy; indeed several distributions are right-skewed (e.g. MonthlyIncome DistanceFromHome, YearsAtCompany). Data transformation methods may be required to approach a normal distribution prior to fitting a model to the data.
2. Age distribution is a slightly right-skewed normal distribution with the bulk of the staff between 25 and 34 years old.

Feature distribution by target attribute

Age

[26]:
```
plt.figure(figsize=(15,6))
plt.style.use('seaborn-colorblind')
plt.grid(True, alpha=0.5)
sns.kdeplot(df.loc[df['Attrition'] == 'No', 'Age'], label = 'Active Employee')
sns.kdeplot(df.loc[df['Attrition'] == 'Yes', 'Age'], label = 'Ex-Employees')
plt.xlim(left=18, right=60)
plt.xlabel('Age (years)')
plt.ylabel('Density')
plt.legend()
plt.title('Age Distribution in Percent by Attrition Status');
```



2.3.2 Categorical Features

[27]:
```
# Creating a function to plot histograms
def barplot(i):
    fig = px.histogram(Attrition, x = Attrition[i], template = 'plotly_dark',
                       title = f'Attrition in {i}', text_auto = 'd3-format')
    fig.show()
```

[28]:
```
# Creating visualizations for categorical values
barplot('Business_Travel')
barplot('Department')
barplot('Education_Field')
barplot('Gender')
barplot('Job_Role')
barplot('Marital_Status')
```

EDA Conclusions

Categorical Variables

1. When analyzing categorical variables, we can see that most employees who have left worked for the Research & Development department, with most of them being laboratory technicians, sales executives or research scientists.

2. Most of them had a Bachelor's degree and their education field was mostly either Life Sciences, Medical and Marketing.

3. How can we make the work environment better? What kind of changes must be done, especially for the research and development personel? These are important questions to be asked.

Numerical Variables

1. Looking at the attrition per age histogram, It's noticeable that as employees grow old, the less they tend to leave, and most of our employee attrition is made of employees ranging from 25 to 35 years old. The data also shows that the more working years, more years at the company, and more years in current role employees accumulate, the less likely they are to leave.

2. Those who've less percent salary hike tend to leave more than those with a higher percentual hike in salary.

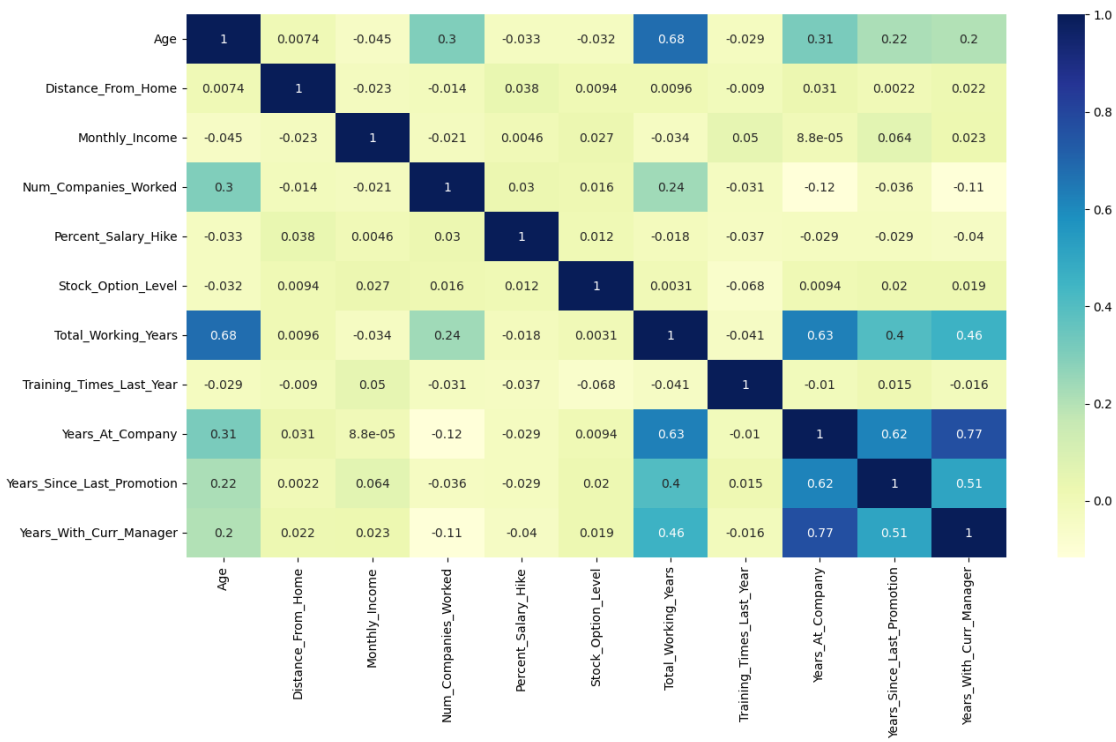
3. So mostly, employees who leave tend to be young, with less time working in the company and at the beginning of their career in general, since most of these employees were working for less than 10 years in total.

4. It's also curious to see that a lot of these employees had less than 2 years working with their last manager. Could we be having issues with some managers? How well are they trained to deal with their teams and the people they led? Are we, as a company doing the best we can to assure a good relationship between managers and teams?

2.3.3 MultiVariate Analysis

[29]:
```
ax = plt.subplots(figsize=(15, 8))
sns.heatmap(data= df.corr(),annot=True,cmap="YlGnBu")
```

[29]: <AxesSubplot:>



As shown above, "YearsWithCurrentManager", "YearsSinceLastPromotion" and "YearsAtCompany" are positively correlated to Attrition;

while "Total Working Years", "Job Level", and "PercentSalaryHike" are negatively correlated to Attrition.

2.4 Section 04: KPI/ Metric based questions

[30]:
```
#creating 10 different buckets for Monthly_Income (Deciles)
df['salary_bins'] = pd.qcut(df['Monthly_Income'],14)
df.head()
```

[30]:
```
   Age Attrition    Business_Travel              Department  \
0   51        No      Travel_Rarely                   Sales
1   31       Yes  Travel_Frequently  Research & Development
2   32        No  Travel_Frequently  Research & Development
3   38        No         Non-Travel  Research & Development
4   32        No      Travel_Rarely  Research & Development

   Distance_From_Home Education Education_Field Employee_ID  Gender Job_Level  \
0                   6         2   Life Sciences           1  Female         1
1                  10         1   Life Sciences           2  Female         1
2                  17         4           Other           3    Male         4
3                   2         5   Life Sciences           4    Male         3
4                  10         1         Medical           5    Male         1

   … Monthly_Income Num_Companies_Worked  Percent_Salary_Hike  \
0  …         131160                    1                   11
1  …          41890                    0                   23
2  …         193280                    1                   15
3  …          83210                    3                   11
4  …          23420                    4                   12

   Stock_Option_Level  Total_Working_Years  Training_Times_Last_Year  \
0                   0                    1                         6
1                   1                    6                         3
2                   3                    5                         2
3                   3                   13                         5
4                   2                    9                         2

   Years_At_Company  Years_Since_Last_Promotion  Years_With_Curr_Manager  \
0                 1                           0                        0
1                 5                           1                        4
2                 5                           0                        3
3                 8                           7                        5
4                 6                           0                        4

               salary_bins
```

```
0  (114294.286, 167990.0]
1    (39040.0, 44021.429]
2    (167990.0, 199990.0]
3    (73858.571, 95820.0]
4      (21940.0, 24390.0]

[5 rows x 22 columns]
```

[31]:
```python
#create empty data frame
KPI = pd.DataFrame()
print(KPI)
```

```
Empty DataFrame
Columns: []
Index: []
```

[32]:
```python
# calculating Average of Monthly_Income for every bin
KPI['Average'] = df.groupby("salary_bins").Monthly_Income.mean().round()
KPI.head(14)
```

[32]:

salary_bins	Average
(10089.999, 21940.0]	18828.0
(21940.0, 24390.0]	23297.0
(24390.0, 27430.0]	26058.0
(27430.0, 31720.0]	29243.0
(31720.0, 39040.0]	35265.0
(39040.0, 44021.429]	41643.0
(44021.429, 49190.0]	46589.0
(49190.0, 54670.0]	52117.0
(54670.0, 62727.143]	58396.0
(62727.143, 73858.571]	66917.0
(73858.571, 95820.0]	83667.0
(95820.0, 114294.286]	103484.0
(114294.286, 167990.0]	140179.0
(167990.0, 199990.0]	185642.0

[33]:
```python
# calculating minimum value of Monthly_Income for every bin
KPI['Minimum_Value'] = df.groupby("salary_bins").Monthly_Income.min()
KPI.head(14)
```

[33]:

salary_bins	Average	Minimum_Value
(10089.999, 21940.0]	18828.0	10090
(21940.0, 24390.0]	23297.0	22010
(24390.0, 27430.0]	26058.0	24400
(27430.0, 31720.0]	29243.0	27560

```
(31720.0, 39040.0]       35265.0          31800
(39040.0, 44021.429]     41643.0          39070
(44021.429, 49190.0]     46589.0          44030
(49190.0, 54670.0]       52117.0          49300
(54670.0, 62727.143]     58396.0          54680
(62727.143, 73858.571]   66917.0          62740
(73858.571, 95820.0]     83667.0          74030
(95820.0, 114294.286]   103484.0          96020
(114294.286, 167990.0]  140179.0         115100
(167990.0, 199990.0]    185642.0         168230
```

[34]:
```python
# calculating maximum value of Monthly_Income for every bin
KPI['Maximum_Value'] = df.groupby("salary_bins").Monthly_Income.max()
KPI.head(14)
```

[34]:
```
                         Average  Minimum_Value  Maximum_Value
salary_bins
(10089.999, 21940.0]     18828.0          10090          21940
(21940.0, 24390.0]       23297.0          22010          24390
(24390.0, 27430.0]       26058.0          24400          27430
(27430.0, 31720.0]       29243.0          27560          31720
(31720.0, 39040.0]       35265.0          31800          39040
(39040.0, 44021.429]     41643.0          39070          44010
(44021.429, 49190.0]     46589.0          44030          49080
(49190.0, 54670.0]       52117.0          49300          54670
(54670.0, 62727.143]     58396.0          54680          62720
(62727.143, 73858.571]   66917.0          62740          73790
(73858.571, 95820.0]     83667.0          74030          95820
(95820.0, 114294.286]   103484.0          96020         114160
(114294.286, 167990.0]  140179.0         115100         167990
(167990.0, 199990.0]    185642.0         168230         199990
```

[35]:
```python
# calculating no of employees for every bin
KPI['No_Of_Emp'] = df.groupby("salary_bins").size()
KPI.head(14)
```

[35]:
```
                         Average  Minimum_Value  Maximum_Value  No_Of_Emp
salary_bins
(10089.999, 21940.0]     18828.0          10090          21940        315
(21940.0, 24390.0]       23297.0          22010          24390        312
(24390.0, 27430.0]       26058.0          24400          27430        313
(27430.0, 31720.0]       29243.0          27560          31720        313
(31720.0, 39040.0]       35265.0          31800          39040        317
(39040.0, 44021.429]     41643.0          39070          44010        308
(44021.429, 49190.0]     46589.0          44030          49080        313
(49190.0, 54670.0]       52117.0          49300          54670        314
(54670.0, 62727.143]     58396.0          54680          62720        312
```

```
(62727.143, 73858.571]   66917.0          62740          73790        313
(73858.571, 95820.0]     83667.0          74030          95820        315
(95820.0, 114294.286]   103484.0          96020         114160        311
(114294.286, 167990.0]  140179.0         115100         167990        314
(167990.0, 199990.0]    185642.0         168230         199990        312
```

[36]:
```python
# calculating no of employees attrition for every bin
KPI['No_of_people_Attrition'] = df.loc[df['Attrition'] == 'Yes'].
 ↪groupby("salary_bins").size()
KPI.head(14)
```

[36]:
```
                         Average  Minimum_Value  Maximum_Value  No_Of_Emp  \
salary_bins
(10089.999, 21940.0]     18828.0          10090          21940        315
(21940.0, 24390.0]       23297.0          22010          24390        312
(24390.0, 27430.0]       26058.0          24400          27430        313
(27430.0, 31720.0]       29243.0          27560          31720        313
(31720.0, 39040.0]       35265.0          31800          39040        317
(39040.0, 44021.429]     41643.0          39070          44010        308
(44021.429, 49190.0]     46589.0          44030          49080        313
(49190.0, 54670.0]       52117.0          49300          54670        314
(54670.0, 62727.143]     58396.0          54680          62720        312
(62727.143, 73858.571]   66917.0          62740          73790        313
(73858.571, 95820.0]     83667.0          74030          95820        315
(95820.0, 114294.286]   103484.0          96020         114160        311
(114294.286, 167990.0]  140179.0         115100         167990        314
(167990.0, 199990.0]    185642.0         168230         199990        312

                        No_of_people_Attrition
salary_bins
(10089.999, 21940.0]                        57
(21940.0, 24390.0]                          41
(24390.0, 27430.0]                          65
(27430.0, 31720.0]                          47
(31720.0, 39040.0]                          51
(39040.0, 44021.429]                        48
(44021.429, 49190.0]                        44
(49190.0, 54670.0]                          48
(54670.0, 62727.143]                        77
(62727.143, 73858.571]                      54
(73858.571, 95820.0]                        45
(95820.0, 114294.286]                       47
(114294.286, 167990.0]                      39
(167990.0, 199990.0]                        42
```

[37]:
```python
# calculating percent of employees were attrition for every bin
KPI['Attrition rate'] = KPI['No_of_people_Attrition']/KPI['No_Of_Emp'] *100
```

```
KPI.head(14)
```

[37]:

```
                          Average  Minimum_Value  Maximum_Value  No_Of_Emp  \
salary_bins
(10089.999, 21940.0]      18828.0          10090          21940        315
(21940.0, 24390.0]        23297.0          22010          24390        312
(24390.0, 27430.0]        26058.0          24400          27430        313
(27430.0, 31720.0]        29243.0          27560          31720        313
(31720.0, 39040.0]        35265.0          31800          39040        317
(39040.0, 44021.429]      41643.0          39070          44010        308
(44021.429, 49190.0]      46589.0          44030          49080        313
(49190.0, 54670.0]        52117.0          49300          54670        314
(54670.0, 62727.143]      58396.0          54680          62720        312
(62727.143, 73858.571]    66917.0          62740          73790        313
(73858.571, 95820.0]      83667.0          74030          95820        315
(95820.0, 114294.286]    103484.0          96020         114160        311
(114294.286, 167990.0]   140179.0         115100         167990        314
(167990.0, 199990.0]     185642.0         168230         199990        312

                         No_of_people_Attrition  Attrition rate
salary_bins
(10089.999, 21940.0]                         57       18.095238
(21940.0, 24390.0]                           41       13.141026
(24390.0, 27430.0]                           65       20.766773
(27430.0, 31720.0]                           47       15.015974
(31720.0, 39040.0]                           51       16.088328
(39040.0, 44021.429]                         48       15.584416
(44021.429, 49190.0]                         44       14.057508
(49190.0, 54670.0]                           48       15.286624
(54670.0, 62727.143]                         77       24.679487
(62727.143, 73858.571]                       54       17.252396
(73858.571, 95820.0]                         45       14.285714
(95820.0, 114294.286]                        47       15.112540
(114294.286, 167990.0]                       39       12.420382
(167990.0, 199990.0]                         42       13.461538
```

[38]:

```
# rounding attrition rate by 2 decimals
KPI['Attrition rate'] = KPI['Attrition rate'].round(2)
KPI.head(14)
```

[38]:

```
                          Average  Minimum_Value  Maximum_Value  No_Of_Emp  \
salary_bins
(10089.999, 21940.0]      18828.0          10090          21940        315
(21940.0, 24390.0]        23297.0          22010          24390        312
(24390.0, 27430.0]        26058.0          24400          27430        313
(27430.0, 31720.0]        29243.0          27560          31720        313
(31720.0, 39040.0]        35265.0          31800          39040        317
```

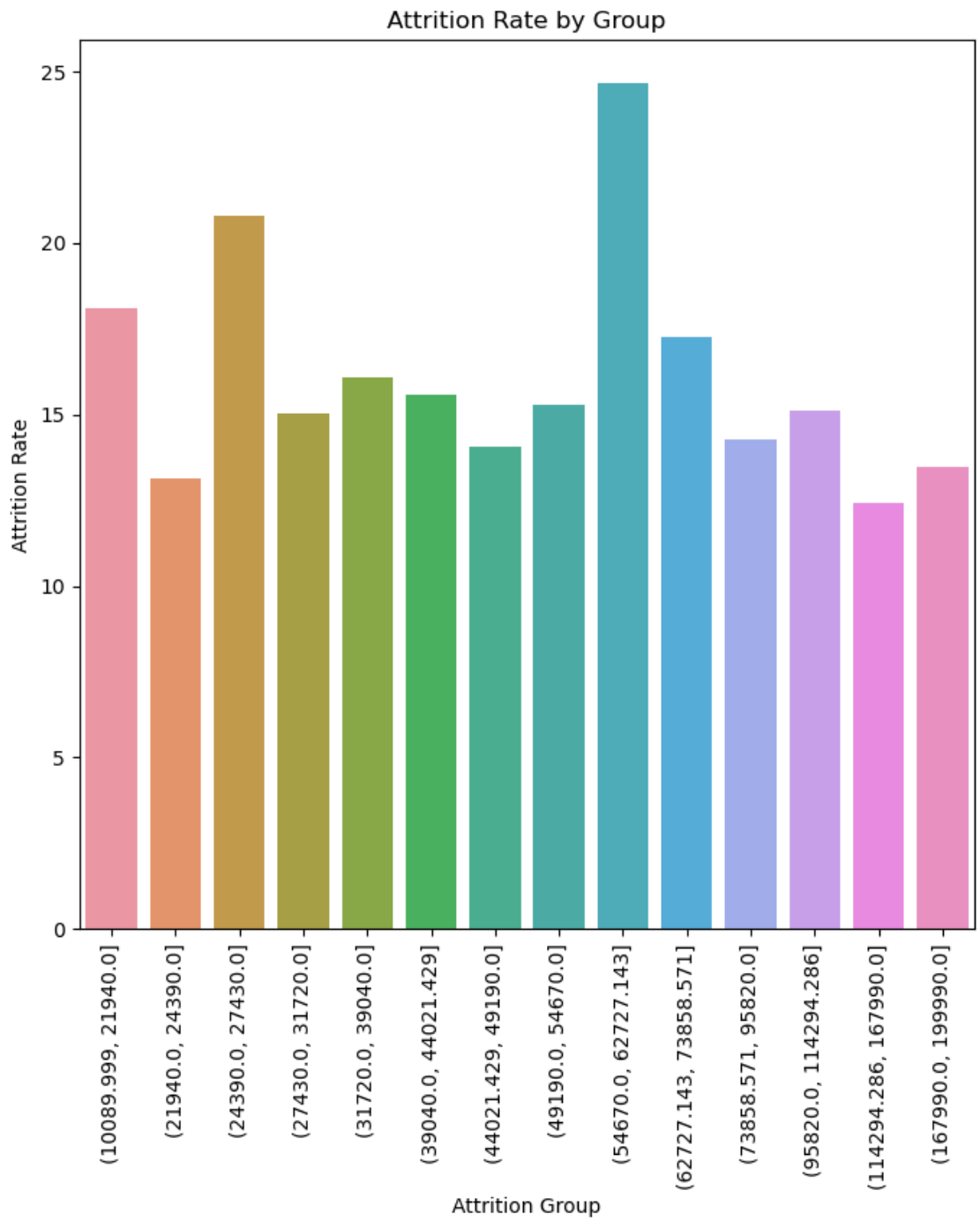
```
(39040.0, 44021.429]     41643.0          39070          44010        308
(44021.429, 49190.0]     46589.0          44030          49080        313
(49190.0, 54670.0]       52117.0          49300          54670        314
(54670.0, 62727.143]     58396.0          54680          62720        312
(62727.143, 73858.571]   66917.0          62740          73790        313
(73858.571, 95820.0]     83667.0          74030          95820        315
(95820.0, 114294.286]   103484.0          96020         114160        311
(114294.286, 167990.0]  140179.0         115100         167990        314
(167990.0, 199990.0]    185642.0         168230         199990        312

                        No_of_people_Attrition  Attrition rate
salary_bins
(10089.999, 21940.0]                        57           18.10
(21940.0, 24390.0]                          41           13.14
(24390.0, 27430.0]                          65           20.77
(27430.0, 31720.0]                          47           15.02
(31720.0, 39040.0]                          51           16.09
(39040.0, 44021.429]                        48           15.58
(44021.429, 49190.0]                        44           14.06
(49190.0, 54670.0]                          48           15.29
(54670.0, 62727.143]                        77           24.68
(62727.143, 73858.571]                      54           17.25
(73858.571, 95820.0]                        45           14.29
(95820.0, 114294.286]                       47           15.11
(114294.286, 167990.0]                      39           12.42
(167990.0, 199990.0]                        42           13.46
```

[39]:
```
KPI = KPI.reset_index()
KPI.head(14)
```

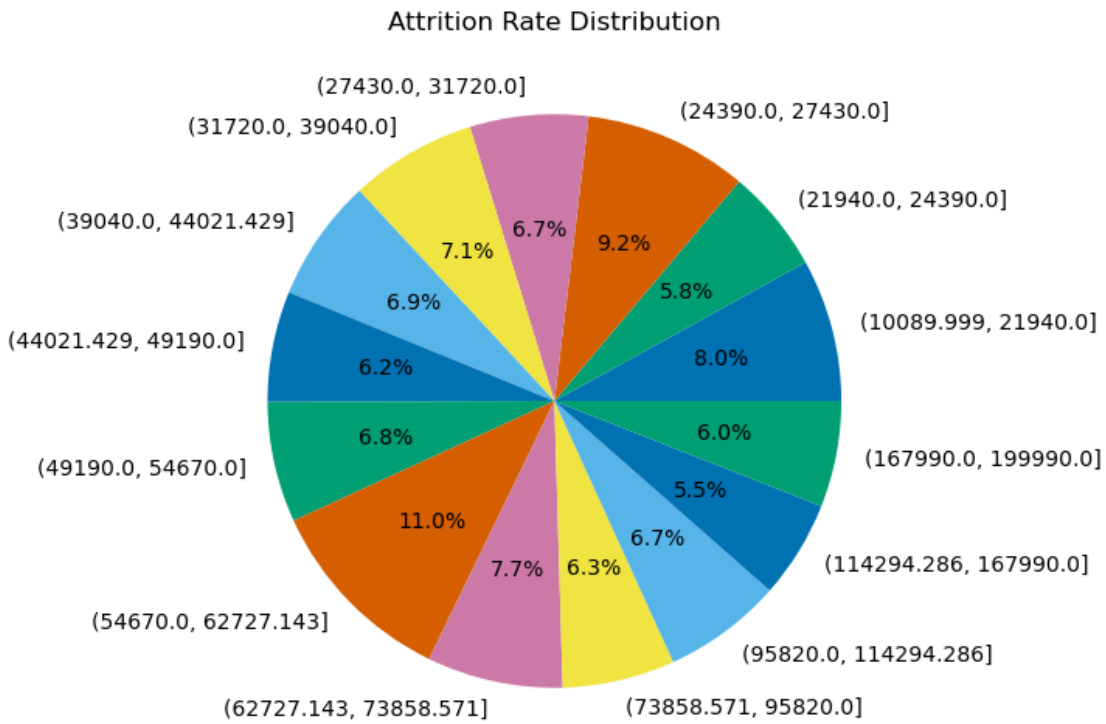
[39]:
```
               salary_bins   Average  Minimum_Value  Maximum_Value  No_Of_Emp  \
0     (10089.999, 21940.0]   18828.0          10090          21940        315
1       (21940.0, 24390.0]   23297.0          22010          24390        312
2       (24390.0, 27430.0]   26058.0          24400          27430        313
3       (27430.0, 31720.0]   29243.0          27560          31720        313
4       (31720.0, 39040.0]   35265.0          31800          39040        317
5     (39040.0, 44021.429]   41643.0          39070          44010        308
6     (44021.429, 49190.0]   46589.0          44030          49080        313
7       (49190.0, 54670.0]   52117.0          49300          54670        314
8     (54670.0, 62727.143]   58396.0          54680          62720        312
9   (62727.143, 73858.571]   66917.0          62740          73790        313
10    (73858.571, 95820.0]   83667.0          74030          95820        315
11   (95820.0, 114294.286]  103484.0          96020         114160        311
12  (114294.286, 167990.0]  140179.0         115100         167990        314
13    (167990.0, 199990.0]  185642.0         168230         199990        312

    No_of_people_Attrition  Attrition rate
```

```
0                        57          18.10
1                        41          13.14
2                        65          20.77
3                        47          15.02
4                        51          16.09
5                        48          15.58
6                        44          14.06
7                        48          15.29
8                        77          24.68
9                        54          17.25
10                       45          14.29
11                       47          15.11
12                       39          12.42
13                       42          13.46
```

[40]:
```
plt.figure(figsize=(8, 8))
ax = sns.barplot(x='salary_bins', y='Attrition rate', data=KPI)
plt.title('Attrition Rate by Group')
plt.xlabel('Attrition Group')
plt.ylabel('Attrition Rate')
plt.xticks(rotation=90)


plt.show()
```



[41]:
```
# Pie Chart
plt.figure(figsize=(6, 6))
plt.pie(KPI['Attrition rate'], labels=KPI['salary_bins'], autopct='%1.1f%%')
plt.title('Attrition Rate Distribution')
plt.show()
```



2.4.1 A few observations can be made based on the Bar chat and pie for numerical features:

-> As we can observe that the for bin (54670.0, 62727.143) attrition rate was more i.e., 11.0 % compared to other bins. follows attrition rate more for the bins (21940.0, 24390.0) = 9.2%, (10089.999, 21940.0] = 8.0%, (62727.143, 73858.571) = 7.7 and as follows

-> As we can observe that we cannot get a conclusion on binning we need further investigation.

[42]:
```
df.head()
```

[42]:
```
   Age Attrition    Business_Travel              Department  \
0   51        No      Travel_Rarely                   Sales
1   31       Yes  Travel_Frequently  Research & Development
2   32        No  Travel_Frequently  Research & Development
3   38        No         Non-Travel  Research & Development
4   32        No      Travel_Rarely  Research & Development

   Distance_From_Home Education Education_Field Employee_ID  Gender Job_Level  \
0                   6         2   Life Sciences           1  Female         1
1                  10         1   Life Sciences           2  Female         1
2                  17         4           Other           3    Male         4
3                   2         5   Life Sciences           4    Male         3
```

```
4                   10         1         Medical           5    Male         1

   … Monthly_Income Num_Companies_Worked  Percent_Salary_Hike  \
0  …         131160                    1                   11
1  …          41890                    0                   23
2  …         193280                    1                   15
3  …          83210                    3                   11
4  …          23420                    4                   12

   Stock_Option_Level  Total_Working_Years  Training_Times_Last_Year  \
0                   0                    1                         6
1                   1                    6                         3
2                   3                    5                         2
3                   3                   13                         5
4                   2                    9                         2

   Years_At_Company  Years_Since_Last_Promotion  Years_With_Curr_Manager  \
0                 1                           0                        0
1                 5                           1                        4
2                 5                           0                        3
3                 8                           7                        5
4                 6                           0                        4

              salary_bins
0  (114294.286, 167990.0]
1    (39040.0, 44021.429]
2    (167990.0, 199990.0]
3    (73858.571, 95820.0]
4      (21940.0, 24390.0]

[5 rows x 22 columns]
```

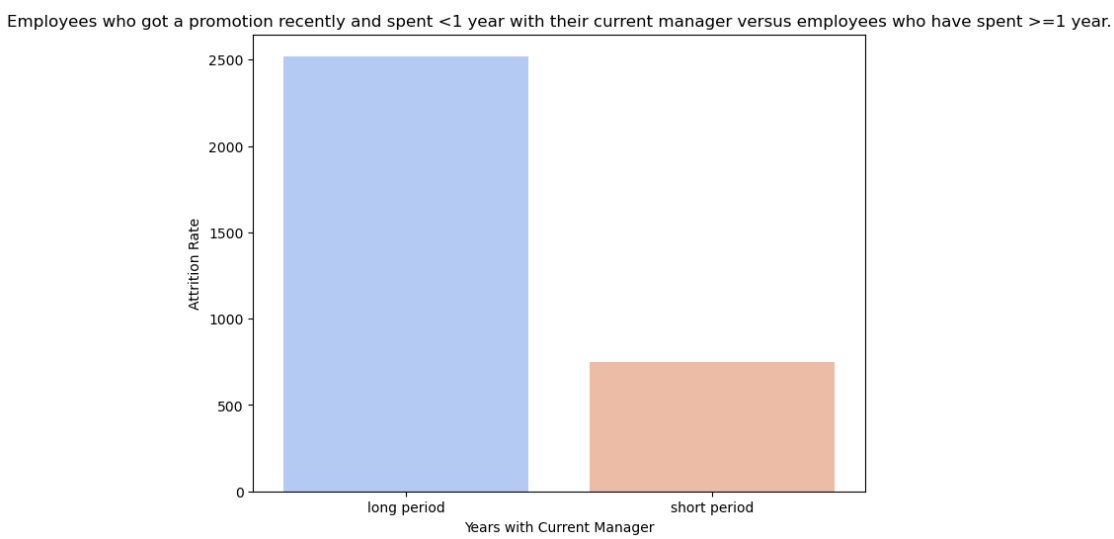
2.5 analyzing Attrition rate for a combination of Two variables

[43]:
```
## Encoding the Attrition Column for Modelling
df['Attrition'] = [1 if i == 'Yes' else 0 for i in df['Attrition']]
```

2.5.1 Employees who got a promotion recently and spent <1 year with their current manager versus employees who have spent >=1 year.

[44]:
```
# creating column Emp_promot_rectly,Years_With_Curr_Manager_cat as per the␣
↪requirement

df['Emp_promot_rectly'] = df['Years_Since_Last_Promotion'].apply(lambda x:␣
↪'prom _recently' if x <= 2 else 'not_recent')
```

```
df['Years_With_Curr_Manager_cat'] = df['Years_With_Curr_Manager'].apply(lambda␣
↪x: 'short period' if x < 1 else 'long period')

# Filter the data to focus on employees who got a promotion recently
filtered_data = df[df['Emp_promot_rectly'] == 'prom _recently']

# Group by Manager_Years_Category and calculate the count of employees in each␣
↪group
promotion_counts =filtered_data['Years_With_Curr_Manager_cat'].value_counts().
↪reset_index()

# Rename the columns for the plot
promotion_counts.columns = ['Manager_Years_Category', 'Count']

# Plot the bar chart
plt.figure(figsize=(8, 6))
sns.barplot(x='Manager_Years_Category', y='Count', data=promotion_counts,␣
↪palette='coolwarm')

# Customize the plot
plt.title("Employees who got a promotion recently and spent <1 year with their␣
↪current manager versus employees who have spent >=1 year.")
plt.xlabel("Years with Current Manager")
plt.ylabel("Attrition Rate")


# Show the plot
plt.show()
```



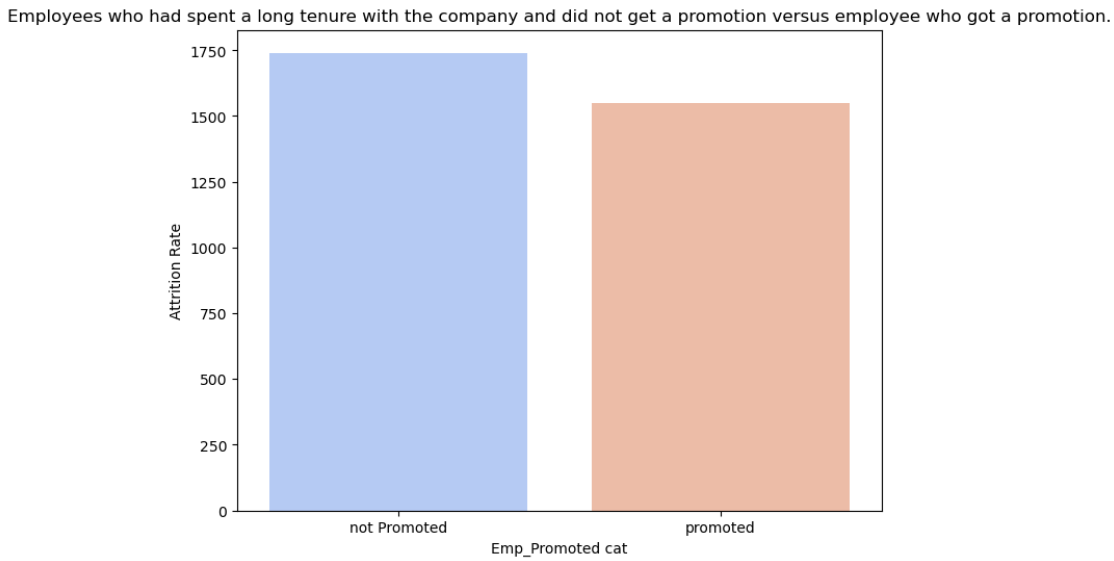
2.6 Employees who got a Promotion, but salary increment was low versus employee who got a higher increment.

[45]:
```
# creating column Emp_Promoted,salary_increment as per the requirement

df['Emp_Promoted'] = df['Years_Since_Last_Promotion'].apply(lambda x:␣
 ↪'promoted' if x < 1 else 'not Promoted')

df['salary_increment'] = df['Percent_Salary_Hike'].apply(lambda x:␣
 ↪'low_increment' if x < 20 else 'high_increment')


# Filter the data to focus on employees who got a promoted
filtered_data_1 = df[df['Emp_Promoted'] == 'promoted']


# Group by salary_increment and calculate the count of employees in each group
promotion_counts_1 =filtered_data_1['salary_increment'].value_counts().
 ↪reset_index()

# Rename the columns for the plot
promotion_counts_1.columns = ['salary increment cat', 'Count']

# Plot the bar chart
plt.figure(figsize=(8, 6))
sns.barplot(x='salary increment cat', y='Count', data=promotion_counts_1,␣
 ↪palette='coolwarm')


# Customize the plot
plt.title("Employees who got a Promotion, but salary increment was low versus␣
 ↪employee who got a higher increment.")
plt.xlabel("increment_rate")
plt.ylabel("Attrition Rate")
# Show the plot
plt.show()
```

2.7 Employees who had spent a long tenure with the company and did not get a promotion versus employee who got a promotion.

[46]:
```
# creating column pent a long tenure as per the requirement

df['spent a long tenure'] = df['Years_At_Company'].apply(lambda x: 'spent_long'␣
↪if x < 10 else 'spent_short')

# Filter the data to focus on employees who got a spent a long tenure
filtered_data_2 = df[df['spent a long tenure'] == 'spent_long']


# Group by Emp_Promoted and calculate the count of employees in each group
promotion_counts_2 = filtered_data_2['Emp_Promoted'].value_counts().
↪reset_index()

# Rename the columns for the plot
promotion_counts_2.columns = ['Emp_Promoted cat', 'Count']

# Plot the bar chart
plt.figure(figsize=(8, 6))
sns.barplot(x='Emp_Promoted cat', y='Count', data=promotion_counts_2,␣
↪palette='coolwarm')
```

```
# Customize the plot
plt.title("Employees who had spent a long tenure with the company and did not␣
↪get a promotion versus employee who got a promotion.")
plt.xlabel("Emp_Promoted cat")
plt.ylabel("Attrition Rate")
# Show the plot
plt.show()
```



2.8 Employees who had spent a long tenure with the company and got a low salary increment versus employee who got a decent hike.

[47]:
```
# Group by salary_increment and calculate the count of employees in each group
promotion_counts_3 = filtered_data_2['salary_increment'].value_counts().
↪reset_index()

# Rename the columns for the plot
promotion_counts_3.columns = ['salary_increment', 'Count']


# Plot the bar chart
plt.figure(figsize=(8, 6))
sns.barplot(x='salary_increment', y='Count', data=promotion_counts_3,␣
↪palette='coolwarm')


# Customize the plot
plt.title("Employees who had spent a long tenure with the company and got a low␣
↪salary increment versus employee who got a decent hike.")
```

```
plt.xlabel("salary_increment")
plt.ylabel("Attrition Rate")
# Show the plot
plt.show()
```

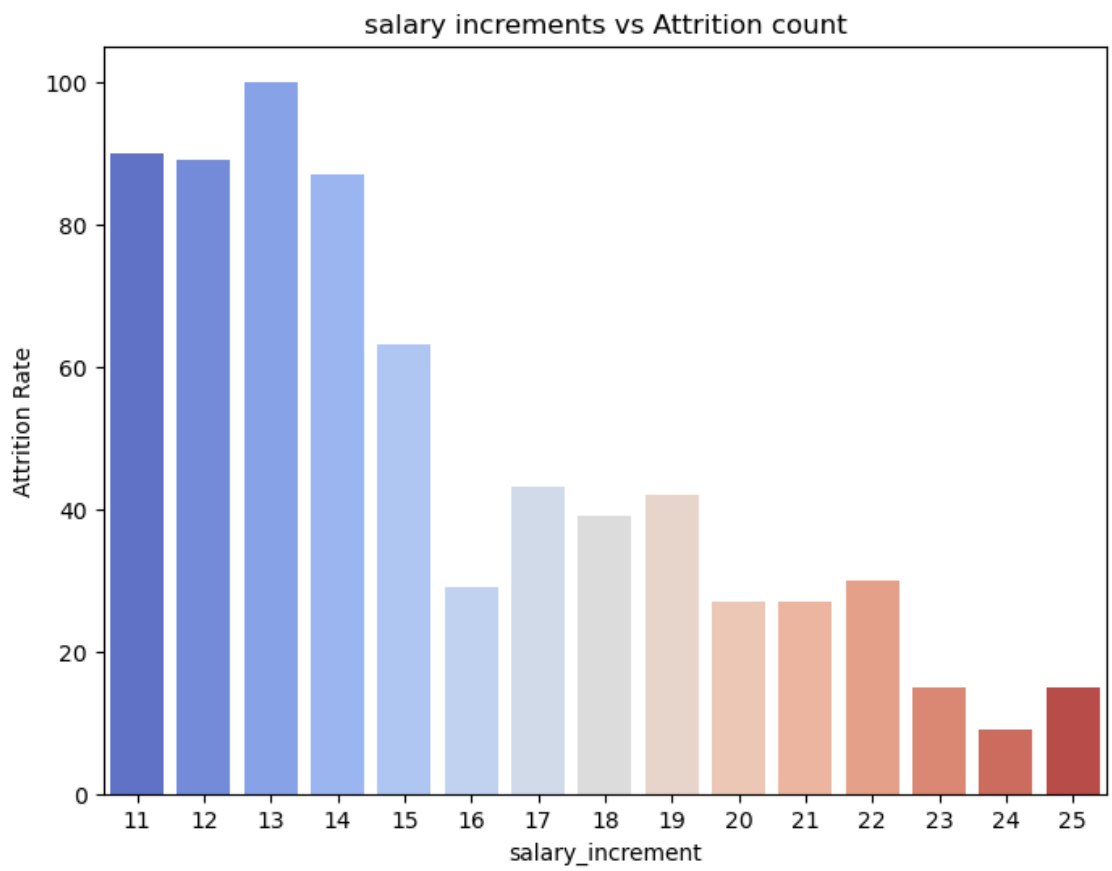
2.9 Section 05: Open-ended questions and recommendations

2.9.1 Which of the below are strong drivers of Attrition:

2.9.2 1) Low salary increments

[48]:
```
# Filter the data to focus on employees who left the company
filtered_data_5 = df[df['Attrition'] == 1]


# Group by Percent_Salary_Hike and calculate the count of employees who where␣
↪left company in each group
promotion_counts_5 = filtered_data_5['Percent_Salary_Hike'].value_counts().
↪reset_index()

# Rename the columns for the plot
promotion_counts_5.columns = ['Percent_Salary_Hike', 'Count']

# Plot the bar chart
plt.figure(figsize=(8, 6))
sns.barplot(x='Percent_Salary_Hike', y='Count', data=promotion_counts_5,␣
↪palette='coolwarm')
```

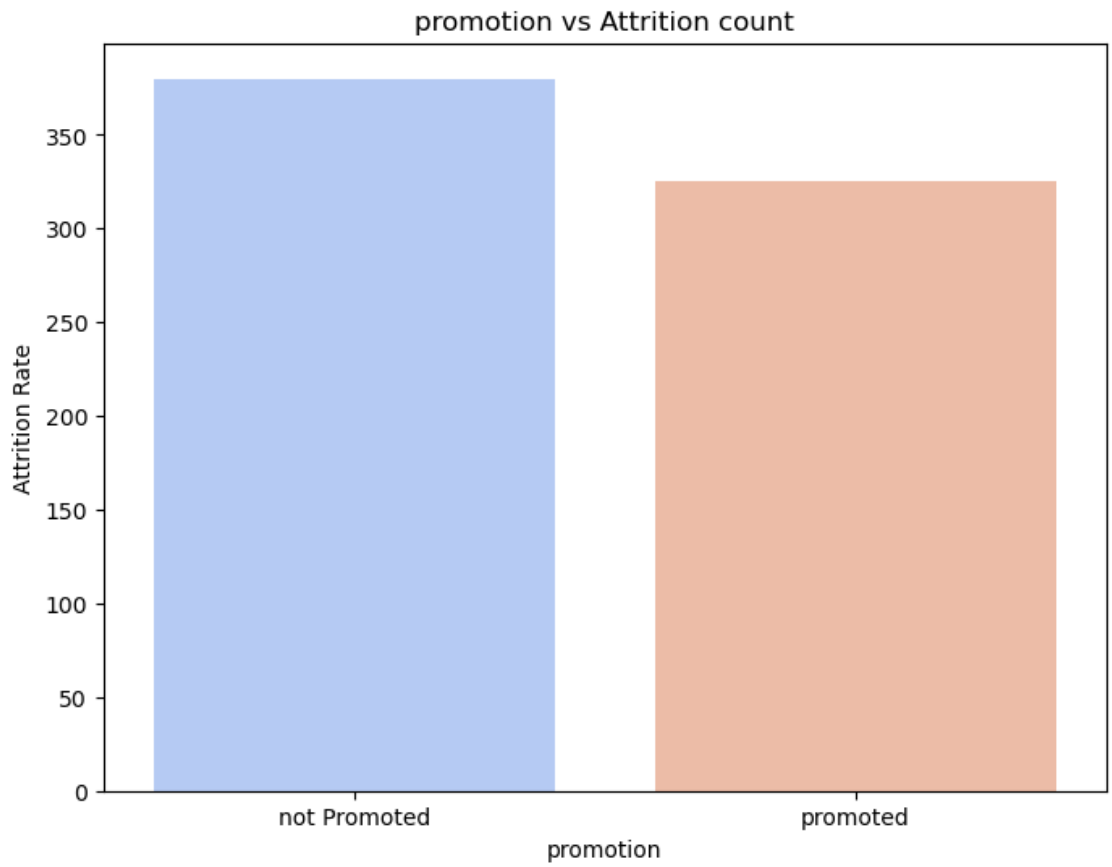
```
# Customize the plot
plt.title("salary increments vs Attrition count")
plt.xlabel("salary_increment")
plt.ylabel("Attrition Rate")
# Show the plot
plt.show()
```



2.9.3 2) No promotion

[49]:
```
# Group by Emp_Promoted and calculate the count of employees who where left␣
↪company in each group
promotion_counts_6 = filtered_data_5['Emp_Promoted'].value_counts().
↪reset_index()

# Rename the columns for the plot
promotion_counts_6.columns = ['Emp_Promoted', 'Count']

# Plot the bar chart
plt.figure(figsize=(8, 6))
```

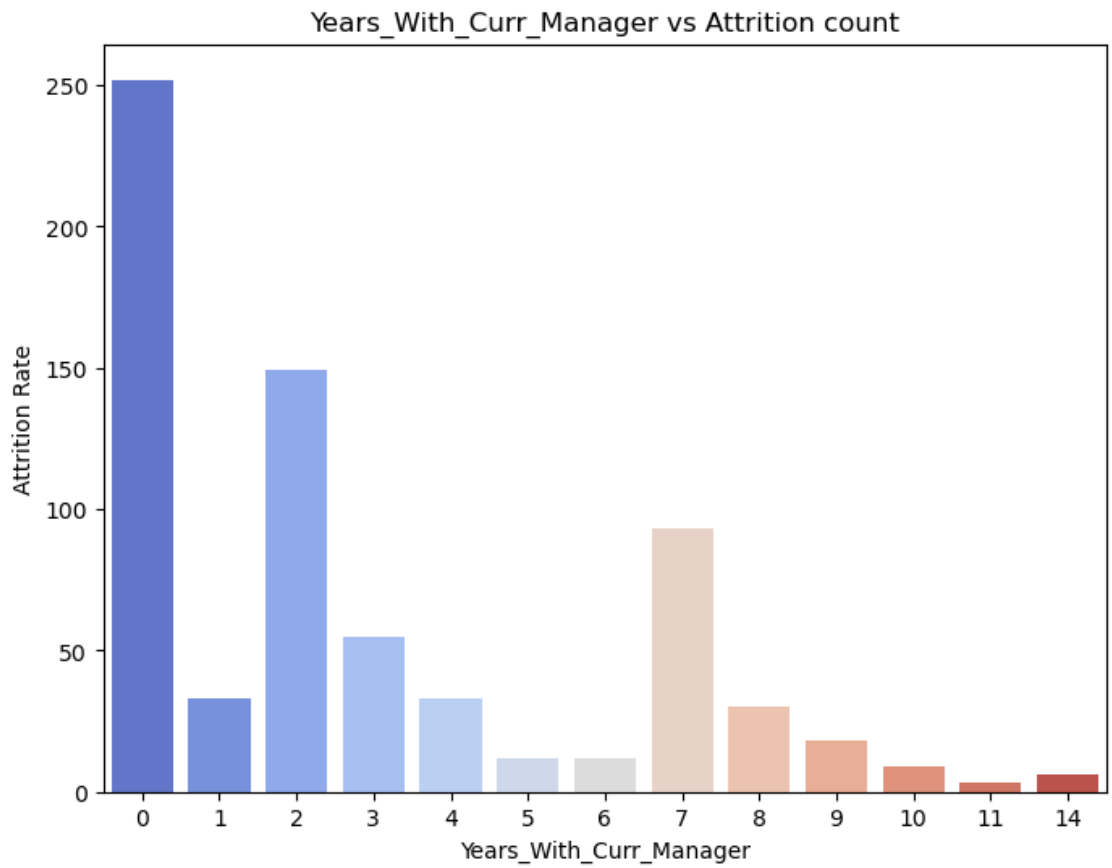
```
sns.barplot(x='Emp_Promoted', y='Count', data=promotion_counts_6,␣
↪palette='coolwarm')


# Customize the plot
plt.title("promotion vs Attrition count")
plt.xlabel("promotion")
plt.ylabel("Attrition Rate")
# Show the plot
plt.show()
```



2.9.4 3) Current manager

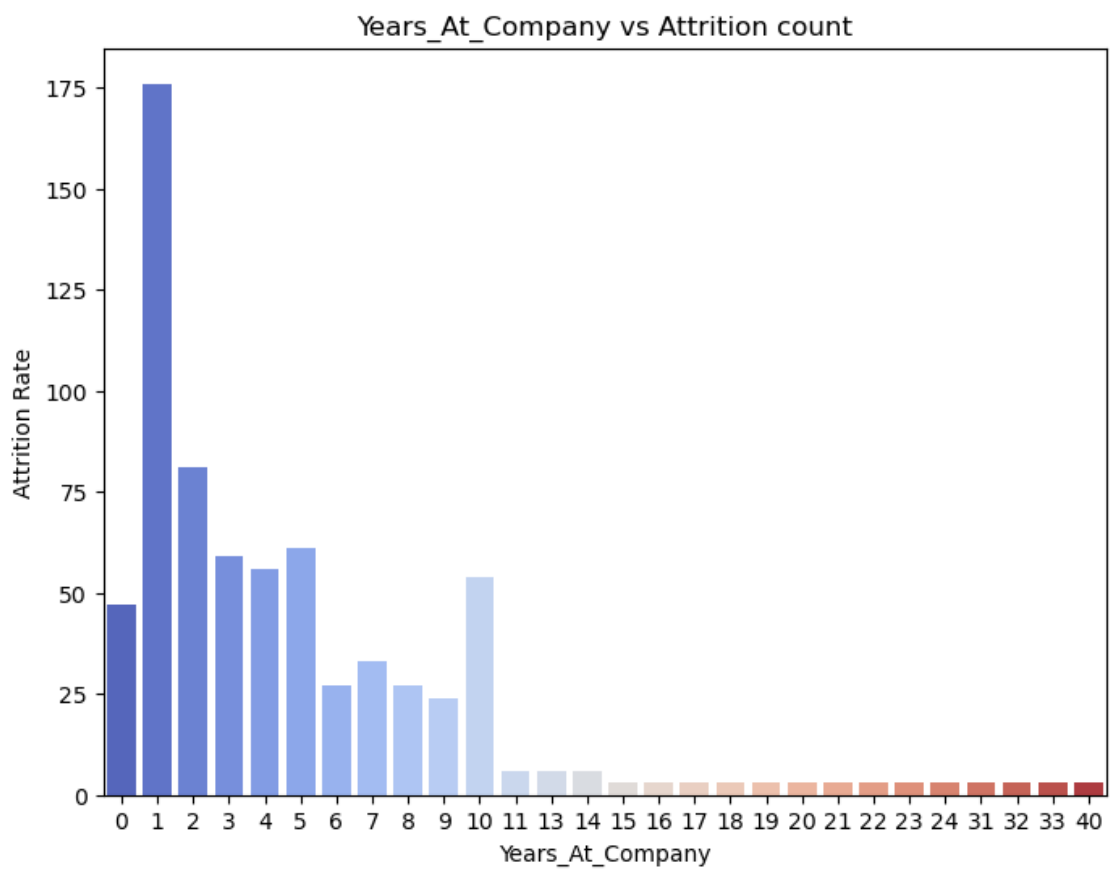
[50]:
```
# Group by Years_With_Curr_Manager and calculate the count of employees who␣
↪where left company in each group
promotion_counts_7 = filtered_data_5['Years_With_Curr_Manager'].value_counts().
↪reset_index()

# Rename the columns for the plot
```

```
promotion_counts_7.columns = ['Years_With_Curr_Manager', 'Count']

# Plot the bar chart
plt.figure(figsize=(8, 6))
sns.barplot(x='Years_With_Curr_Manager', y='Count', data=promotion_counts_7,␣
 ↪palette='coolwarm')


# Customize the plot
plt.title("Years_With_Curr_Manager vs Attrition count")
plt.xlabel("Years_With_Curr_Manager")
plt.ylabel("Attrition Rate")
# Show the plot
plt.show()
```



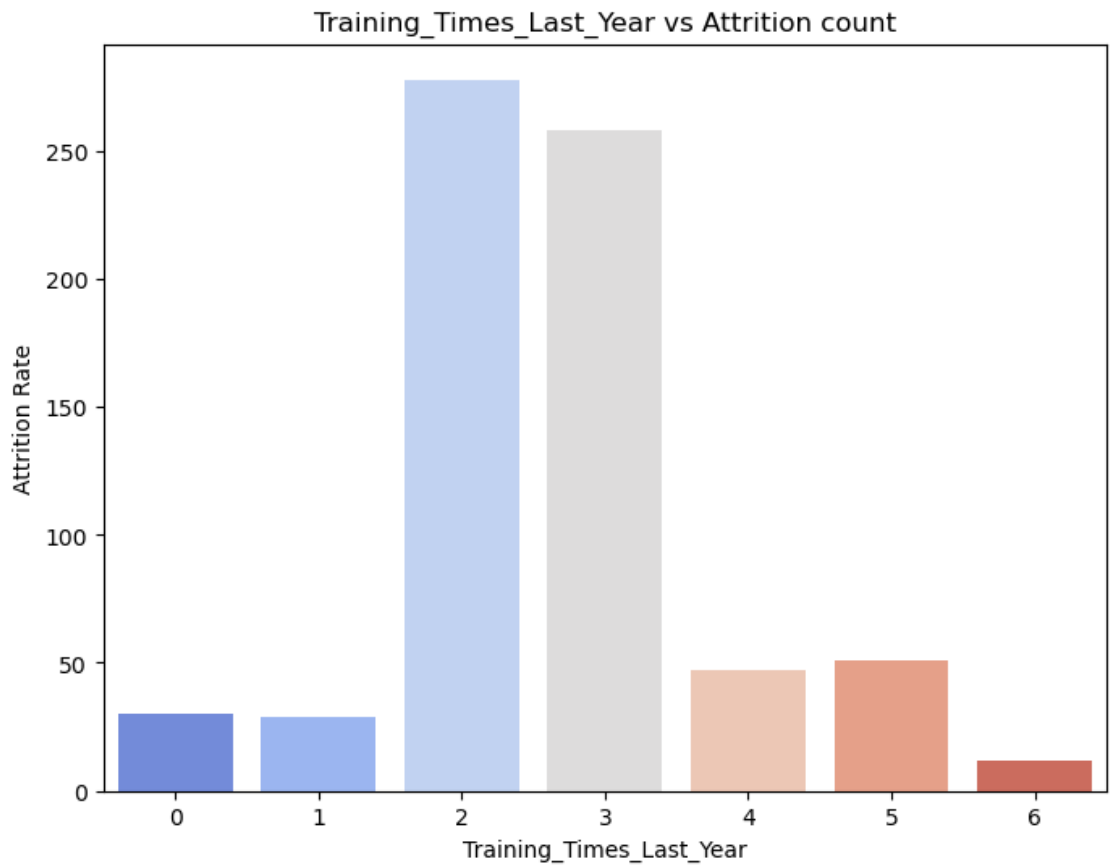
2.9.5 4) Need for a change (after having served the company for long)

[51]:
```
# Group by Years_At_Company and calculate the count of employees who where left␣
↪company in each group
promotion_counts_8 = filtered_data_5['Years_At_Company'].value_counts().
↪reset_index()

# Rename the columns for the plot
promotion_counts_8.columns = ['Years_At_Company', 'Count']

# Plot the bar chart
plt.figure(figsize=(8, 6))
sns.barplot(x='Years_At_Company', y='Count', data=promotion_counts_8,␣
↪palette='coolwarm')


# Customize the plot
plt.title("Years_At_Company vs Attrition count")
plt.xlabel("Years_At_Company")
plt.ylabel("Attrition Rate")
# Show the plot
plt.show()
```



2.9.6 5) Lesser training time spent in last year

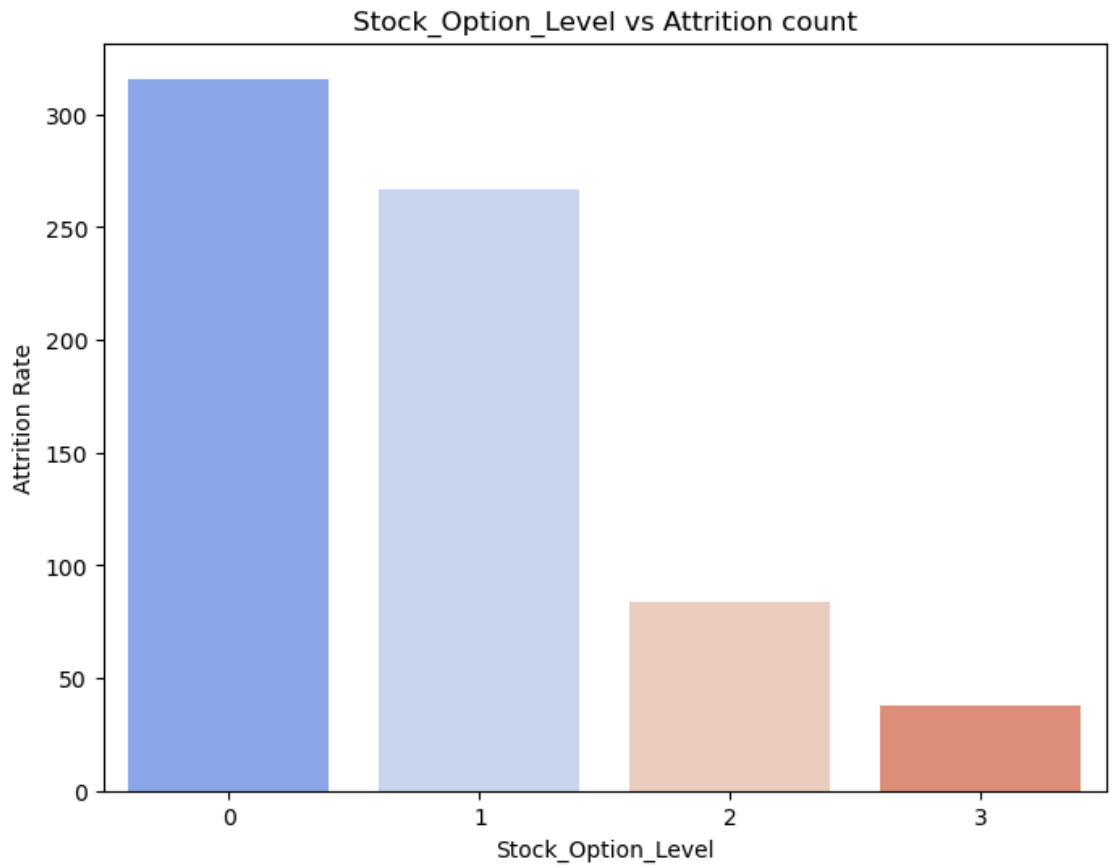
[52]:
```
# Group by Training_Times_Last_Year and calculate the count of employees who␣
↪where left company in each group
promotion_counts_9 = filtered_data_5['Training_Times_Last_Year'].value_counts().
↪reset_index()

# Rename the columns for the plot
promotion_counts_9.columns = ['Training_Times_Last_Year', 'Count']

# Plot the bar chart
plt.figure(figsize=(8, 6))
sns.barplot(x='Training_Times_Last_Year', y='Count', data=promotion_counts_9,␣
↪palette='coolwarm')


# Customize the plot
plt.title("Training_Times_Last_Year vs Attrition count")
plt.xlabel("Training_Times_Last_Year")
plt.ylabel("Attrition Rate")
# Show the plot
plt.show()
```



2.9.7 6)Stock Options not being given

[53]:
```
# Group by Training_Times_Last_Year and calculate the count of employees who␣
↪where left company in each group
promotion_counts_9 = filtered_data_5['Stock_Option_Level'].value_counts().
↪reset_index()

# Rename the columns for the plot
promotion_counts_9.columns = ['Stock_Option_Level', 'Count']

# Plot the bar chart
plt.figure(figsize=(8, 6))
sns.barplot(x='Stock_Option_Level', y='Count', data=promotion_counts_9,␣
↪palette='coolwarm')


# Customize the plot
plt.title("Stock_Option_Level vs Attrition count")
plt.xlabel("Stock_Option_Level")
```

```
plt.ylabel("Attrition Rate")
# Show the plot
plt.show()
```



Always open for feedback and suggestions.If it helps Thumbs Up !!!